6. Solve the equation $x^3 - 9x^2 + 14x + 24 = 0$ if it is given that two of its roots are in the ratio $3:2$.

7. If α, β, and γ are the roots of the polynomial equation $ax^3 + bx^2 + cx + d = 0$, find the value of $\sum \frac{\alpha}{\beta\gamma}$ in terms of the coefficients.

8. If α, β, γ, and δ are the roots of the polynomial equation $2x^4 + 5x^3 - 7x^2 + 8 = 0$, find a quadratic equation with integer coefficients whose roots are $\alpha + \beta + \gamma + \delta$ and $\alpha\beta\gamma\delta$.

9. If p and q are the roots of the equation $lx^2 + nx + n = 0$, show that $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{n}{l}} = 0$.

10. If the equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root, show that it must be equal to $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$.

11. A 12 metre tall tree was broken into two parts. It was found that the height of the part which was left standing was the cube root of the length of the part that was cut away. Formulate this into a mathematical problem to find the height of the part which was left standing.

3.4 Nature of Roots and Nature of Coefficients of Polynomial Equations

3.4.1 Imaginary Roots

For a quadratic equation with real coefficients, if $\alpha + i\beta$ is a root, then $\alpha - i\beta$ is also a root. In this section we shall prove that this is true for higher degree polynomials as well.

We now prove one of the very important theorems in the theory of equations.

Theorem 3.2 (Complex Conjugate Root Theorem)

If a complex number z_0 is a root of a polynomial equation with real coefficients, then its complex conjugate $\overline{z}_0$ is also a root.

Proof

Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_o = 0$ be a polynomial equation with real coefficients. Let z_0 be a root of this polynomial equation. So, $P(z_0) = 0$. Now

$$\begin{aligned} P(\overline{z_0}) &= a_n \overline{z}_0^{\,n} + a_{n-1}\overline{z}_0^{\,n-1} + \cdots + a_1 \overline{z}_0 + a_0 \\ &= a_n \overline{z_0^{\,n}} + a_{n-1}\overline{z_0^{\,n-1}} + \cdots + a_1 \overline{z_0} + a_0 \\ &= \overline{a_n}\,\overline{z_0^{\,n}} + \overline{a_{n-1}}\,\overline{z_0^{\,n-1}} + \cdots + \overline{a_1}\,\overline{z_0} + \overline{a_0} \qquad (a_r = \overline{a_r} \text{ as } a_r \text{ is real for all } r) \\ &= \overline{a_n z_0^{\,n}} + \overline{a_{n-1} z_0^{\,n-1}} + \cdots + \overline{a_1 z_0} + \overline{a_0} \\ &= \overline{a_n z_0^{\,n} + a_{n-1} z_0^{\,n-1} + \cdots + a_1 z_0 + a_0} \quad = \overline{P(z_0)} = \overline{0} = 0. \end{aligned}$$

That is $P(\overline{z}_0) = 0$; this implies that whenever z_0 is a root (i.e. $P(z_0)=0$), its conjugate $\overline{z}_0$ is also a root. ■

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